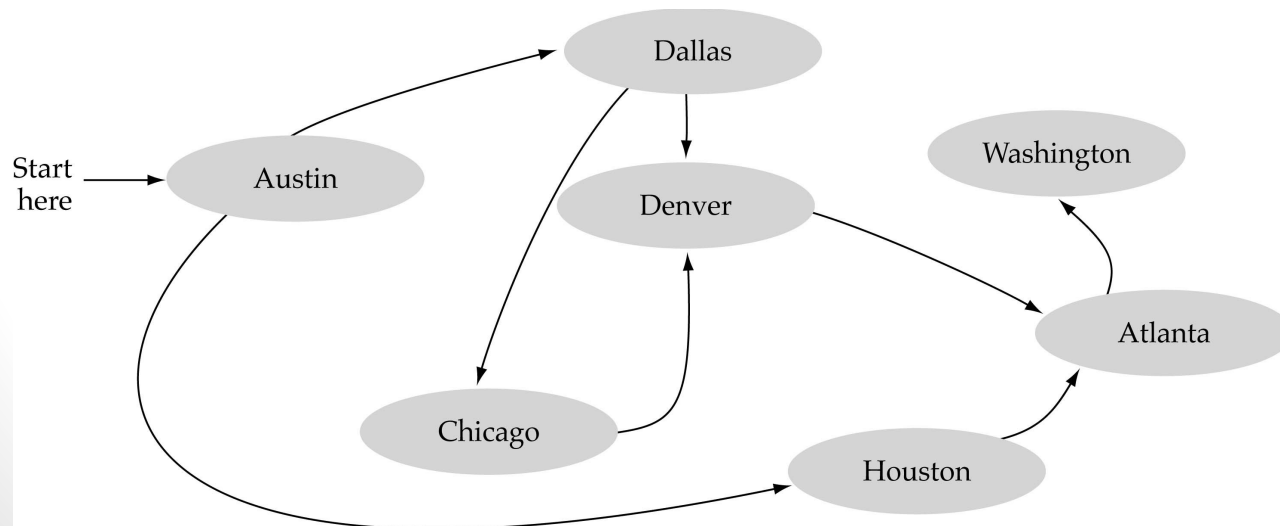


Graph

Afsana Begum
Senior Lecturer,
Software Engineering Department,
Daffodil International University

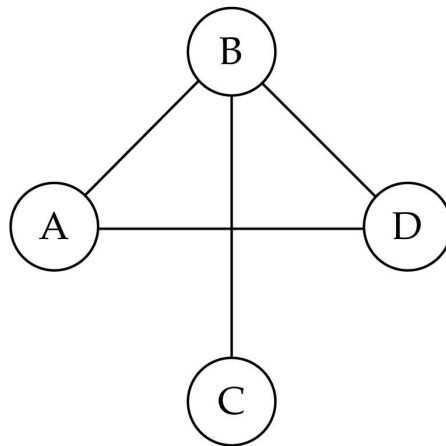
What is a graph?

- A data structure that consists of a set of nodes (*vertices*) and a set of edges that relate the nodes to each other
- The set of edges describes relationships among the vertices



Undirected graphs

- When the edges in a graph have no direction, the graph is called *undirected*



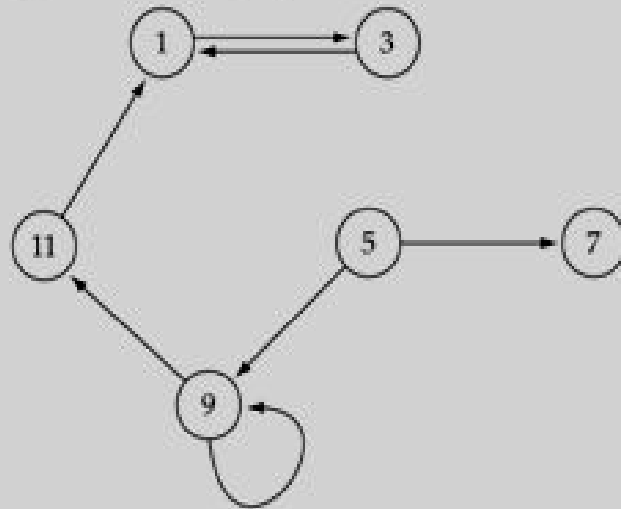
$V(\text{Graph1}) = \{ A, B, C, D \}$

$E(\text{Graph1}) = \{ (A, B), (A, D), (B, C), (B, D) \}$

Directed Graph

- When the edges in a graph have a direction, the graph is called *directed* (or *digraph*)

(b) Graph2 is a directed graph.



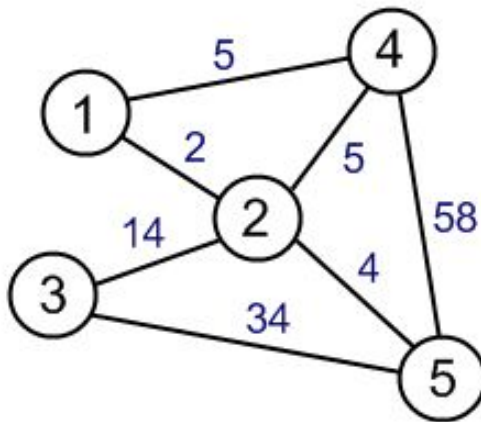
$V(\text{Graph2}) = \{ 1, 3, 5, 7, 9, 11 \}$

$E(\text{Graph2}) = \{(1, 3), (3, 1), (5, 9), (9, 11), (5, 7), (9, 9), (11, 1)\}$

Warning: if the graph is directed, the order of the vertices in each edge is important !!

Weighted graphs

- Here we have the following parts.
- The underlying set for the Vertices set is Integer.
- The underlying set for the weights is Integer.
- The Vertices set = $\{1,2,3,4,5\}$
- The Edge set =
 $\{(1,4,5), (4,5,58), (3,5,34), (2,4,5), (2,5,4), (3,2,14), (1,2,2)\}$



Graph Representation

Vertices can be represent as one dimensional array.

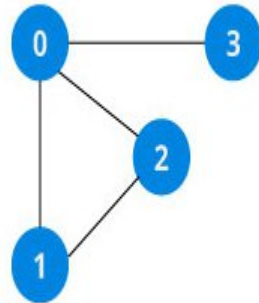
Graph edges are commonly represented in two ways:

1. Adjacency Matrix

An adjacency matrix is 2D array of $V \times V$ vertices. Each row and column represent a vertex.

If the value of any element $a[i][j]$ is 1, it represents that there is an edge connecting vertex i and vertex j .

Adjacency Matrix Example



	0	1	2	3
0	0	1	1	1
1	1	0	1	0
2	1	1	0	0
3	1	0	0	0

Since it is an undirected graph, for edge (0,2), we also need to mark edge (2,0); making the adjacency matrix symmetric about the diagonal. Edge lookup(checking if an edge exists between vertex A and vertex B) is extremely fast in adjacency matrix representation but we have to reserve space for every possible link between all vertices($V \times V$), so it requires more space.

Adjacency Matrix for Flight Connections

graph

.num Vertices

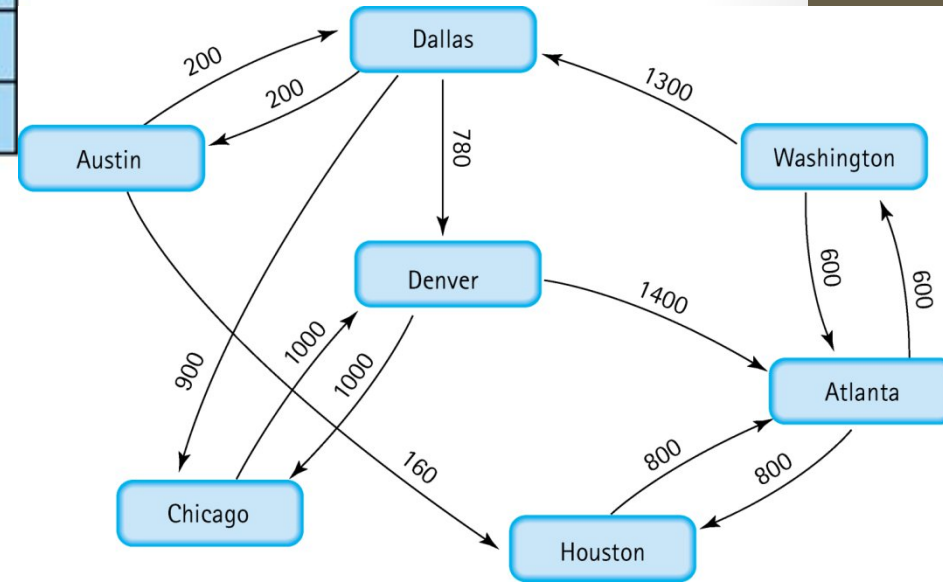
.vertices

[0]	"Atlanta"	"
[1]	"Austin"	"
[2]	"Chicago"	"
[3]	"Dallas"	"
[4]	"Denver"	"
[5]	"Houston"	"
[6]	"Washington"	"
[7]		
[8]		
[9]		

.edges

[0]	0	0	0	0	0	800	600	•	•	•
[1]	0	0	0	200	0	160	0	•	•	•
[2]	0	0	0	0	1000	0	0	•	•	•
[3]	0	200	900	0	780	0	0	•	•	•
[4]	1400	0	1000	0	0	0	0	•	•	•
[5]	800	0	0	0	0	0	0	•	•	•
[6]	600	0	0	1300	0	0	0	•	•	•
[7]	•	•	•	•	•	•	•	•	•	•
[8]	•	•	•	•	•	•	•	•	•	•
[9]	•	•	•	•	•	•	•	•	•	•
	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]

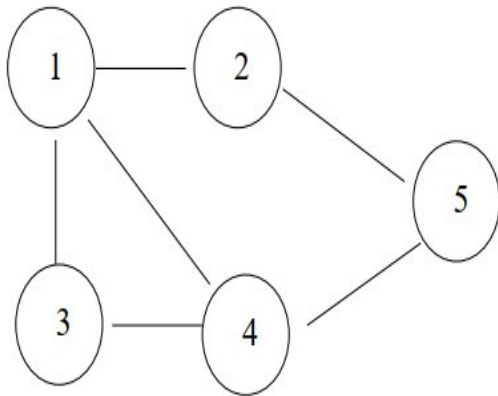
(Array positions marked '•' are undefined)



2. Adjacency List

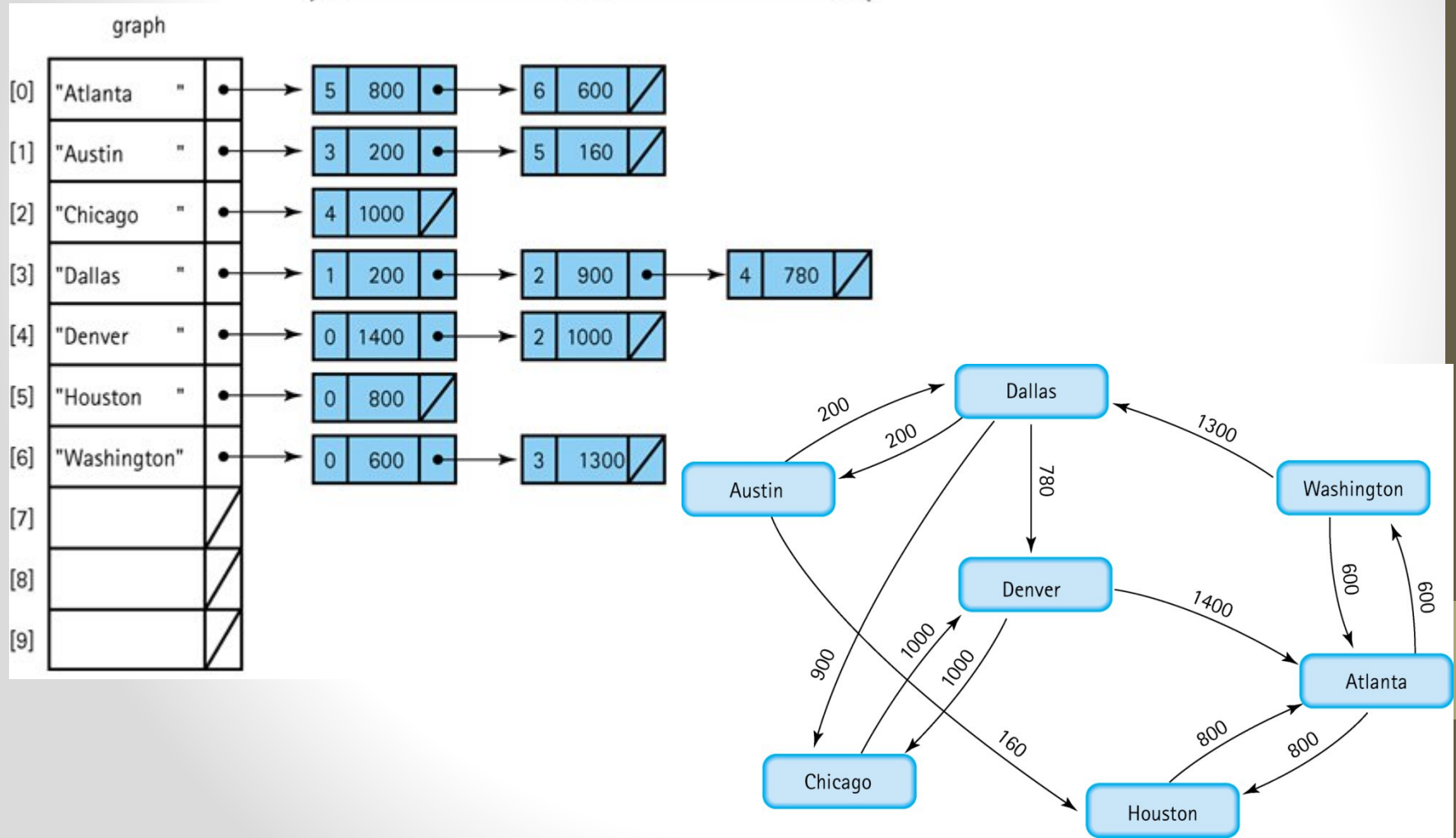
Adjacency list is a collection of unordered lists used to represent a finite graph.

Undirected Adjacency List

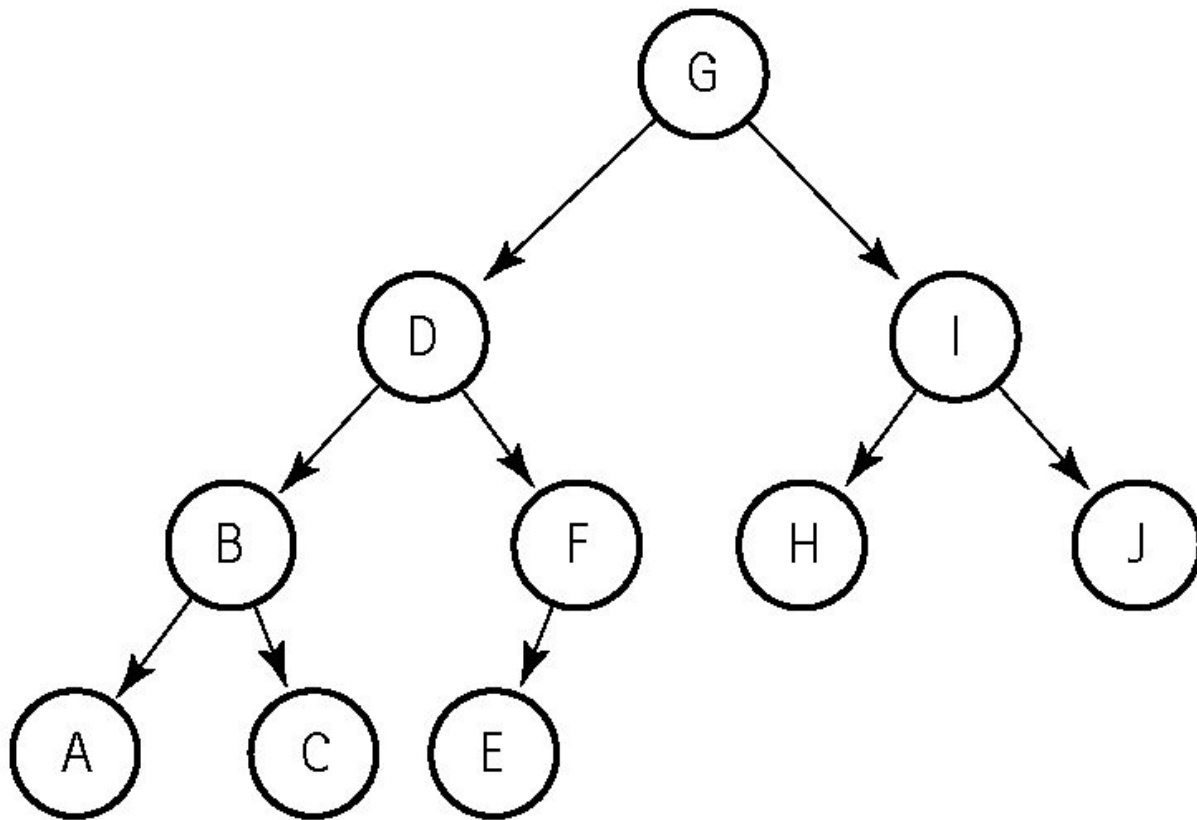


1	→	2	→	3	→	4
2	→	1	→	5		
3	→	4	→	1		
4	→	1	→	3		
5	→	2	→	4		

Adjacency List Representation of Graphs



Trees are special cases of graphs!



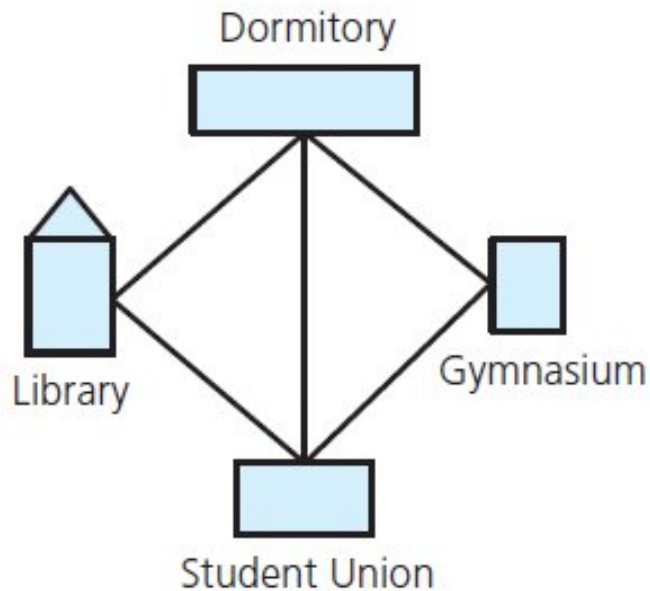
$V(\text{Graph3}) = \{A, B, C, D, E, F, G, H, I, J\}$

$E(\text{Graph3}) = \{(G, D), (G, I), (D, B), (D, F), (I, H), (I, J), (B, A), (B, C), (F, E)\}$

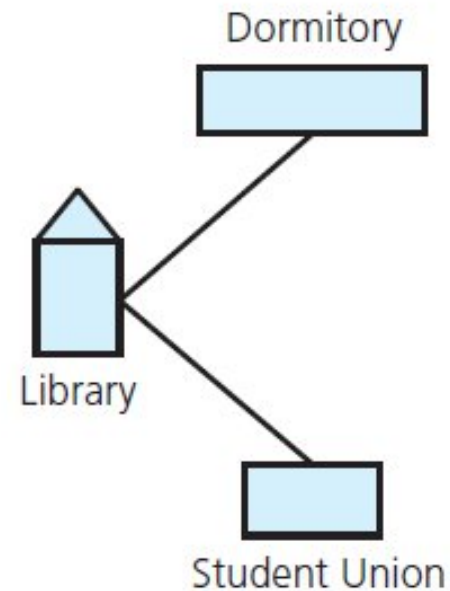
Terminology

- (a) A campus map as a graph;
(b) a subgraph

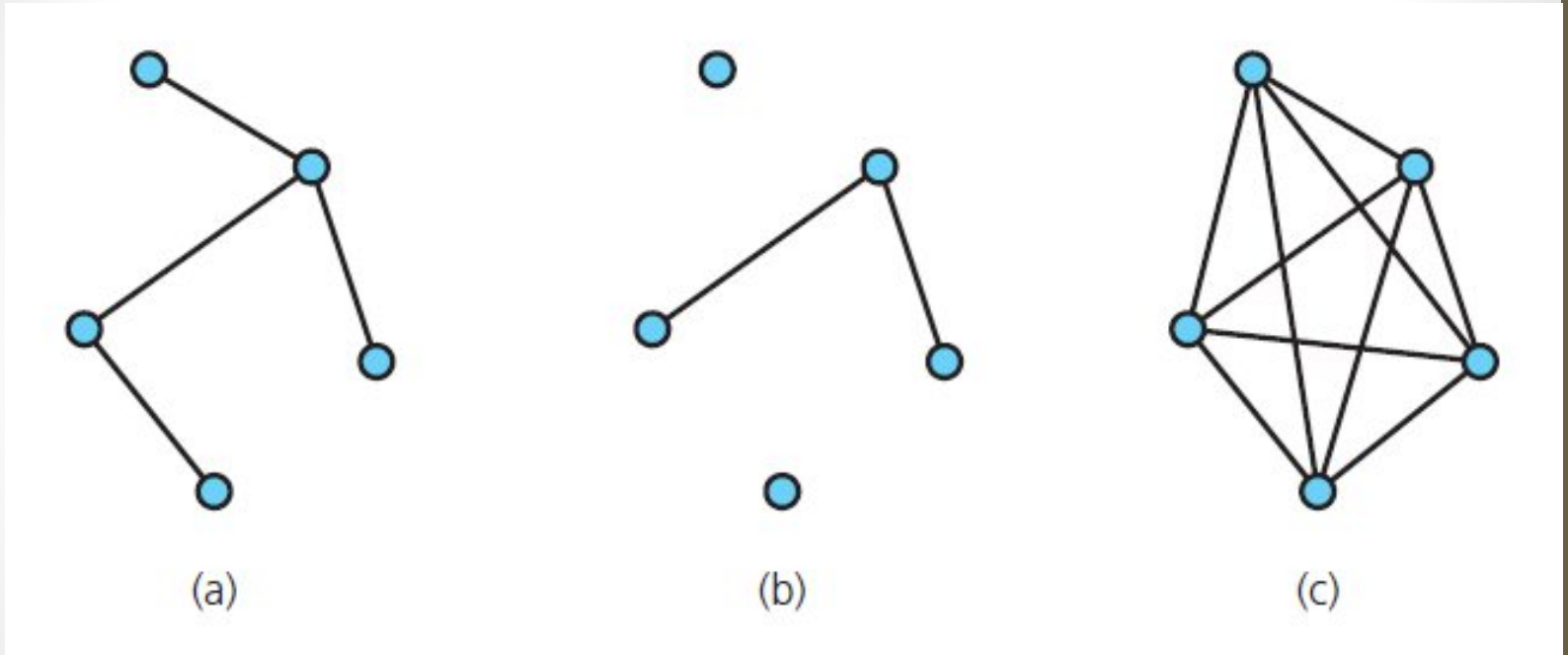
(a)



(b)

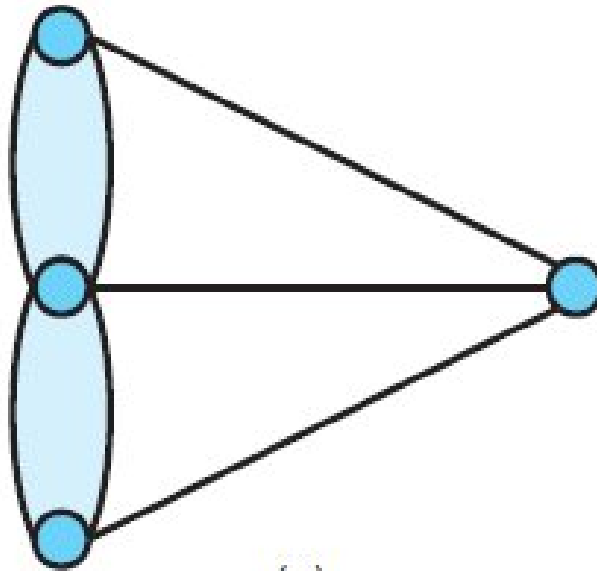


Terminology



- Graphs that are (a) connected; (b) disconnected; and (c) complete

Terminology



(a)

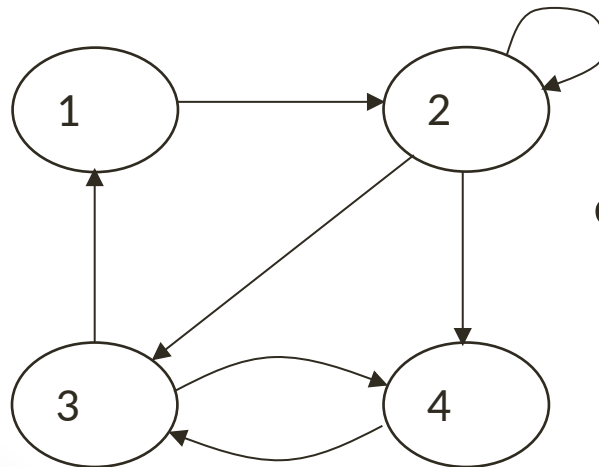


(b)

- (a) A multigraph is not a simple graph;
(b) a self edge is not allowed in a simple graph

Graph terminology

- Path: A sequence of vertices that connects two nodes in a graph
- The length of a path is the number of edges in the path.

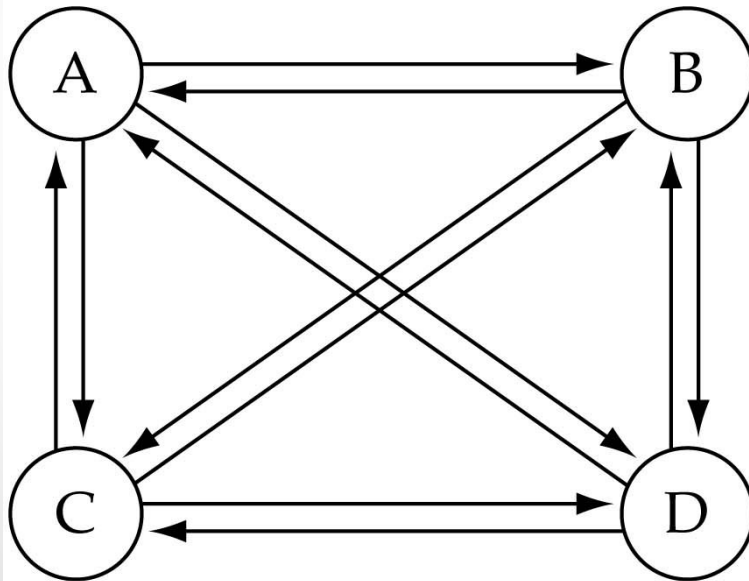


e.g., a path from 1 to 4

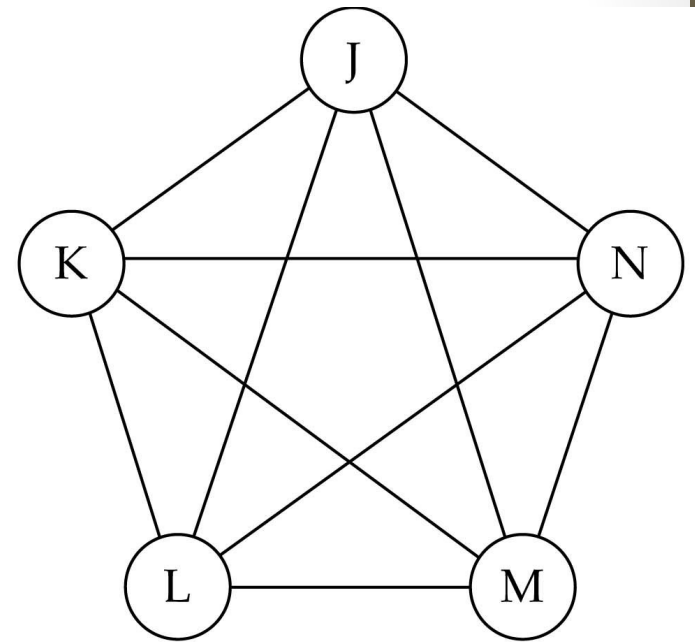
$\langle 1, 2, 3, 4 \rangle$

Graph terminology

Complete graph: A graph in which every vertex is directly connected to every other vertex



(a) Complete directed graph.



(b) Complete undirected graph.

Graph Implementation

Directed Graph:

<https://www.techiedelight.com/implement-graph-data-structure-c/>

Undirected graph:

<https://www.geeksforgeeks.org/graph-and-its-representations/>